

# **The Effects of School Proximity and the School Social Index on Housing Prices**

4th February 2026

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# Agenda

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## 1. Motivation, Existing Literature & Research Question

*Why school proximity and social indices matter.*

## 2. Methodology & Data

*Our empirical model and the data we have used.*

## 3. Results & Main Findings

*Presentation and interpretation of our findings.*

## 4. Conclusion & Limitations

*Concluding thoughts and limitations of our work.*

## **Motivation, Existing Literature & Research Question**

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# Motivation

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- **Personal Interest:** Analyzing the real estate market's reaction to various factors is particularly interesting from an economic standpoint, as it offers insights into market dynamics. Additionally, the topic is personally intriguing, adding an extra layer of motivation to explore it further.
- **Academic and Practical Relevance:** Housing markets are influenced by various social and economic factors, and school quality is also cited as a underestimated determinant of property values, especially in research (Seo and Simons 2009).

# Motivation

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Understanding the relationship between school distance, school quality and property prices can provide valuable insights for researchers and individuals interested in the real estate market.

# Related Literature Background

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Several papers have examined the effect of proximity to schools on housing prices:

- **Empirical Strategies:** While hedonic models are the dominant approach to estimating housing price effects, alternative empirical strategies have also been proposed (see Black and Machin 2011).
- **Findings:** Empirical evidence suggests that proximity to schools positively affects housing prices (Rosiers, Lagana, and Theriault 2001; Huang and Hess 2018).

# Related Literature Background

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- **Other Factors:** Related work also documents capitalization effects of school reputation (Chin and Foong 2006), differences between public and private schools (Sah, Conroy, and Narwold 2016) and school quality (Edusei, Espey, and Lin 2007; Metz 2015).
- **Environment of studies:** Much of the literature focuses on settings with binding school catchment areas and measures school quality primarily through test scores rather than demographic composition.

# Research Question

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**Out contribution:** We study a setting **without binding school catchment areas** (NRW, Germany) and assess whether both school proximity and a **demographically based school social index** are capitalized into housing prices.



# Research Question

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**Research Question:** How do school proximity and school social indices affect the prices of single-family houses, and do these effects differ between primary and secondary schools in North Rhine–Westphalia?

**H1: House prices differ based on the distance to schools**

*Expected: Negative coefficients on distance variables*

**H2: Social index moderates the effect of school proximity on house prices**

*Expected: Better schools leading to stronger price premiums*

## Methodology & Data

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# Empirical Framework

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We estimate regression-based hedonic models to

- examine how school proximity and school characteristics are capitalized into housing prices and
- whether the impact of school proximity on housing prices differs by the school's social index

# Empirical Framework: Hedonic Price Model

$$\log(P_i) = \beta_0 + \beta_1 \text{Dist}_i + X_i' \gamma + \mu_{\ell(i)} + \varepsilon_i$$

- $P_i$  denotes the listing price of house  $i$
- $\text{Dist}_i$  measures the distance to the nearest school (primary or secondary, depending on specification)
- $X_i$  is a vector of housing characteristics
- $\gamma$  is a vector of coefficients corresponding to  $X_i$
- $\mu_{\ell(i)}$  are municipality fixed effects
- $\varepsilon_i$  captures unobserved factors

# Empirical Framework: Distance and Social Index

$$\log(P_i) = \beta_0 + \beta_1 \text{Dist}_i + \beta_2 \text{Index}_i \\ + \beta_3 (\text{Dist}_i \times \text{Index}_i) + X_i' \gamma + \mu_{\ell(i)} + \varepsilon_i$$

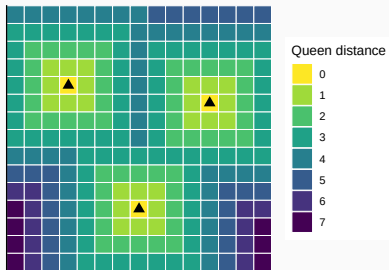
- $P_i$  denotes the listing price of house  $i$
- $\text{Dist}_i$  measures the distance to the nearest school (primary or secondary, depending on specification)
- $X_i$  is a vector of housing characteristics
- $\text{Index}_i$  is the school social index
- $\mu_{\ell(i)}$  are municipality fixed effects
- $\varepsilon_i$  captures unobserved factors

# Empirical Framework: Queen contiguity

$$\text{Dist}_{ij}^Q = \max(|x_i - x_j|, |y_i - y_j|)$$

- Results for distance and the social index are re-estimated using Queen (Chebyshev) distance to check robustness of our results.

Queen distance to nearest school on a 15×15 raster grid



# Data used

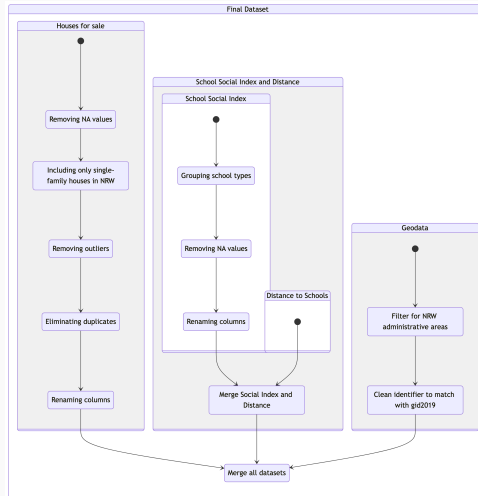
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- **Housing Data:** Cross-section real-estate datasets for Germany published by the Research Data Center Ruhr (FDZ Ruhr) (RWI and ImmobilienScout24 2023).
- **School Data:** School social index dataset from North Rhine-Westphalias Ministry of Schools and Education (Ministerium für Schule und Bildung des Landes Nordrhein-Westfalen 2024).
- **Geodata:** Geodata for Germany's administrative areas used to create graphics of NRW. Taken from Germany's Federal Agency for Cartography and Geodesy (BKG 2025).

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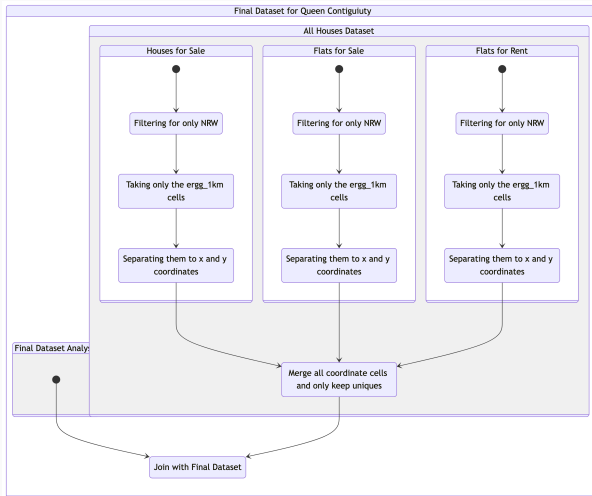
Housing and school data both refer to the year 2022.

# Pre-Processing of Data: Main Analysis





# Pre-Processing of Data: Queen Contiguity



## Results and Main Findings

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# Distance to Schools and House Prices: Hypothesis

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**H1: House prices differ based on the distance to schools**

*Expected: Negative coefficients on distance variables*

## Empirical Framework

- **Outcome:** Log house price
- **Key variables:** Distance to nearest *primary* school, distance to nearest *secondary* school
- **Modeling Strategy:** Continuous, quadratic, and binned specifications

# Distance to Schools and House Prices: Regression Strategy

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## Baseline OLS Controls

### Standardizing housing attributes

- Living area
- Plot area
- Number of rooms
- House age

## Two Distance Designs

### Alternative Specifications

1. Continuous
  - quadratic ( $km^2$ )
2. Binned
  - 0–3 km, 3–6 km, 6–9 km, >9 km

# Regression Specifications

## (1) Naive (Unconditional)

$$\ln(P) = \alpha + \beta_1 D^{prim} + \beta_2 D^{sec} + \varepsilon$$

- **Goal:** Establish raw spatial correlation.
- **Risk:** High Omitted Variable Bias (OVB).

## (2) Baseline (Preferred)

$$\ln(P) = \alpha + \mathbf{D} + \mathbf{X} + \varepsilon$$

- **Controls:** Area, Plot, Rooms, Age.
- **Hypothesis:**  $\beta < 0$  (Negative semi-elasticity).

## (3) Non-Linearity (Polynomial)

$$\ln(P) = \dots + \delta_1 (D^{prim})^2 + \dots$$

- **Test:**  $\delta \neq 0$  implies varying MWTP.
- **Logic:** Test for convex/concave decay.

## (4–5) Robustness (Single Type)

$$\ln(P) = \alpha + \beta_k D^k + \mathbf{X} + \varepsilon$$

- **Idea:** Estimate primary / secondary separately.
- **Check:** Stability against multicollinearity.

# Descriptive Evidence: Price-Distance Gradient

## Negative Slope (H1)

Clear decay in prices as distance increases across specifications.

## Non-Linearity

Steeper drop in the first 3–6 km, flattening afterwards.

## Heterogeneity

**Primary schools** show a visibly steeper gradient than Secondary.

Fig 1. Continuous Trend (Raw Scatter)

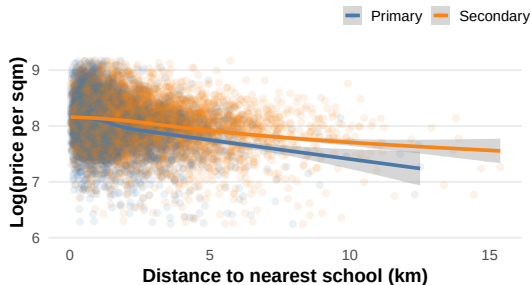
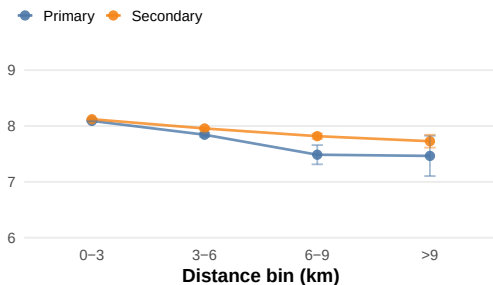


Fig 2. Discrete Gradient (Binned Means)



# Main Results: Continuous Model

| Variable                                | Naive     |         | Baseline  |         | Polynomial |         | Primary Only |         | Secondary Only |         |
|-----------------------------------------|-----------|---------|-----------|---------|------------|---------|--------------|---------|----------------|---------|
|                                         | Coef.     | SE      | Coef.     | SE      | Coef.      | SE      | Coef.        | SE      | Coef.          | SE      |
| Distance to primary (km)                | -0.07***  | (0.007) | -0.071*** | (0.005) | -0.107***  | (0.011) | -0.103***    | (0.005) |                |         |
| Distance to secondary (km)              | -0.031*** | (0.004) | -0.034*** | (0.003) | -0.023***  | (0.008) |              |         | -0.057***      | (0.003) |
| Distance to primary (km) <sup>2</sup>   |           |         |           |         | 0.007***   | (0.002) |              |         |                |         |
| Distance to secondary (km) <sup>2</sup> |           |         |           |         | -0.001     | (0.001) |              |         |                |         |
| N                                       | 5794      |         | 5794      |         | 5794       |         | 5794         |         | 5794           |         |
| R <sup>2</sup>                          | 0.06      |         | 0.436     |         | 0.438      |         | 0.425        |         | 0.419          |         |
| Adj. R <sup>2</sup>                     | 0.06      |         | 0.436     |         | 0.437      |         | 0.425        |         | 0.418          |         |

Note:

Dependent variable: log house price.

HC1 robust standard errors in parentheses.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

# Main Results: Binned Model

| Variable            | Naive     |         | Baseline  |         | Primary Only |         | Secondary Only |         |
|---------------------|-----------|---------|-----------|---------|--------------|---------|----------------|---------|
|                     | Coef.     | SE      | Coef.     | SE      | Coef.        | SE      | Coef.          | SE      |
| Primary: 3-6 km     | -0.147*** | (0.027) | -0.146*** | (0.021) | -0.268***    | (0.019) |                |         |
| Primary: 6-9 km     | -0.361*** | (0.094) | -0.369*** | (0.073) | -0.577***    | (0.071) |                |         |
| Primary: >9 km      | -0.833*** | (0.255) | -0.538*** | (0.199) | -0.793***    | (0.192) |                |         |
| Secondary: 3-6 km   | -0.142*** | (0.017) | -0.146*** | (0.013) |              |         | -0.179***      | (0.012) |
| Secondary: 6-9 km   | -0.241*** | (0.033) | -0.256*** | (0.026) |              |         | -0.331***      | (0.024) |
| Secondary: >9 km    | -0.209*** | (0.08)  | -0.294*** | (0.063) |              |         | -0.435***      | (0.059) |
| N                   | 5794      |         | 5794      |         | 5794         |         | 5794           |         |
| R <sup>2</sup>      | 0.046     |         | 0.42      |         | 0.401        |         | 0.413          |         |
| Adj. R <sup>2</sup> | 0.045     |         | 0.419     |         | 0.4          |         | 0.412          |         |

*Note:*

Dependent variable: log house price. Reference: 0-3 km.

HC1 robust standard errors in parentheses.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$



# Compare: Continuous vs Binned

| Variable                     | Continuous |         | Binned    |         |
|------------------------------|------------|---------|-----------|---------|
|                              | Coef.      | SE      | Coef.     | SE      |
| Distance to primary (km)     | -0.071***  | (0.005) |           |         |
| Distance to secondary (km)   | -0.034***  | (0.003) |           |         |
| Primary: 3-6 km (ref: 0-3)   |            |         | -0.146*** | (0.021) |
| Primary: 6-9 km (ref: 0-3)   |            |         | -0.369*** | (0.073) |
| Primary: >9 km (ref: 0-3)    |            |         | -0.538*** | (0.199) |
| Secondary: 3-6 km (ref: 0-3) |            |         | -0.146*** | (0.013) |
| Secondary: 6-9 km (ref: 0-3) |            |         | -0.256*** | (0.026) |
| Secondary: >9 km (ref: 0-3)  |            |         | -0.294*** | (0.063) |
| N                            | 5794       |         | 5794      |         |
| R <sup>2</sup>               | 0.436      |         | 0.42      |         |
| Adj. R <sup>2</sup>          | 0.436      |         | 0.419     |         |

# Model comparison (main models)

- Continuous fits slightly better.

| Model                 | N    | $R^2$ | Adj. $R^2$ | AIC    | BIC    | CV RMSE | CV MAE | CV $R^2$ |
|-----------------------|------|-------|------------|--------|--------|---------|--------|----------|
| Baseline (Continuous) | 5794 | 0.436 | 0.436      | 4973.8 | 5027.1 | 0.371   | 0.288  | 0.434    |
| Baseline (Binned)     | 5794 | 0.420 | 0.419      | 5143.6 | 5223.6 | 0.377   | 0.293  | 0.416    |

*Note:*

In-sample: higher  $R^2$ /Adj.  $R^2$  is better; lower AIC/BIC is better. Out-of-sample (5-fold CV): lower RMSE/MAE is better; higher  $R^2$  is better.

# Robustness checks (main models)

- Binned is slightly more robust to influential points.

| Model             | BP p-value | Max VIF | Influential ( $D > 4/n$ ) | Max Cook's $D$ |
|-------------------|------------|---------|---------------------------|----------------|
| Good Reference    | <0.001     | 27.57   | 315                       | 0.033          |
| Average Reference | <0.001     | 109.13  | 315                       | 0.033          |
| Bad Reference     | <0.001     | 578.70  | 315                       | 0.033          |

*Note:*

Inference uses textbfHC1 robust SE. Lower BP p-values indicate heteroskedasticity; higher VIF indicates collinearity; larger Cook's  $D$  indicates influential points.

# Robustness check: Queen contiguity (Distance)

| Variable       | Primary Schools |                | Secondary Schools |                |
|----------------|-----------------|----------------|-------------------|----------------|
|                | Coefficient     | Standard Error | Coefficient       | Standard Error |
| Queen distance | -0.077***       | (0.006)        | -0.026***         | (0.003)        |

*Note:*

Dependent variable: log house price, control variables included but not shown.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

# Economic Mechanisms & Interpretations

## 1. Capitalization

*Why closer = expensive?*

- **Convenience Premium:**  
Time savings →  
Capitalized into property values.
- **Sorting:**  
Families cluster → Higher local demand.

## 2. Primary vs Secondary

*Why primary is stronger?*

- **Age Constraints:**  
Young kids need escorting  
(Inelastic demand).
- **Micro-Neighborhoods:**  
Primary schools anchor  
local community ties.

## 3. Non-Linearity

*Why gradient flattens?*

- **Walkability Limit:**  
Premium vanishes after  
**3km.**
- **Substitution:**  
Driving → Public transit at  
longer distances.

# Exploring the Social Index Impact on House Prices

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## H2: Social index moderates the effect of school proximity on house prices

*Expected: Better schools leading to stronger price premiums*

- **Key Finding from Distance Analysis:** We've already established that *households are willing to pay a premium for proximity to schools*
- **Next Question:** Now, we investigate *whether this price premium is driven by school quality, measured through the Social Index*

# What is the Social Index?

**Definition:** The Social Index is a measure that shows the socio-economic and demographic background of a schools students.

**Components of the social index are:**

- Parental Education
- Socio-economic Status of the Family
- Migration Background
- Single-Parent Household
- Unemployment in the Family
- Housing Situation

→ It provides insights into the **social environment** that might influence housing prices in that area.

# Categorization of the Social Index

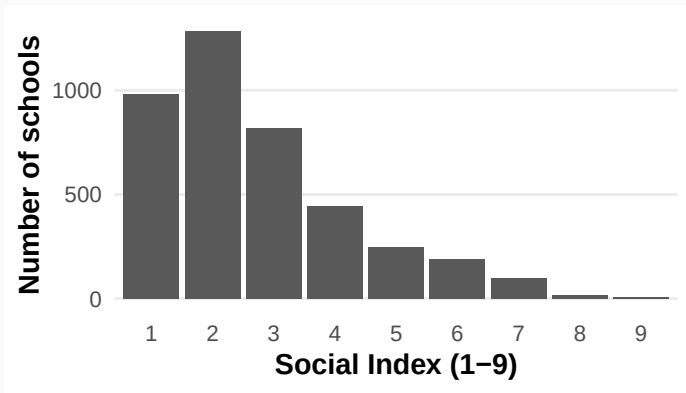
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## Social Index Classification:

- **Good Schools:** Social Index 1-2
- **Average Schools:** Social Index 3-4
- **Bad Schools:** Social Index 5-8
- **Skewed Distribution:** The Social Index classes are heavily skewed, with many schools falling into classes 1, 2 & 3. We excluded class 9 for our analysis because of too less observations



# Categorization of the Social Index: Bar Chart

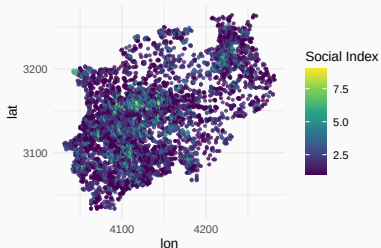


**Graph:** Distribution of Social Index values across schools showing the skewed nature of the data.

# Spatial Distribution of School Social Index in NRW

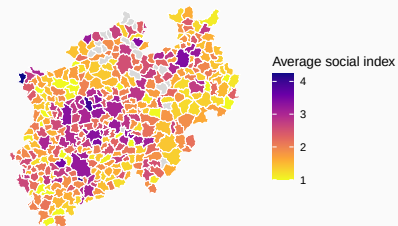
Spatial Distribution of School Social Index in NRW

Each grid cell colored by the social index of the nearest school



Average Social Index by Municipality

Mean household social index within municipalities



# Hedonic Regression Model for Social Index

$$\begin{aligned}\log(\text{price}_i) = & \beta_0 + \beta_1 \text{dist\_primary}_i + \beta_2 \text{dist\_secondary}_i \\ & + \beta_3 \text{school\_quality}_i + \beta_4 (\text{dist\_primary}_i \times \text{school\_quality}_i) \\ & + \beta_5 (\text{dist\_secondary}_i \times \text{school\_quality}_i) + X_i' \gamma + \varepsilon_i\end{aligned}$$

Where:

- $\log(P_i)$  denotes the natural logarithm of the housing price of house  $i$
- $\text{dist}_i^{\text{primary}}$  measures distance to the nearest primary school (km)
- $\text{dist}_i^{\text{secondary}}$  measures distance to the nearest secondary school (km)
- $\text{school\_quality}_i$  is a categorical indicator of school quality
- $X_i$  is a vector of housing characteristics
- $\varepsilon_i$  captures unobserved factors

# Social Index: Regression Results

| Variable                                      | Coefficient | Standard Error |
|-----------------------------------------------|-------------|----------------|
| Distance to primary school (km)               | -0.08***    | (0.007)        |
| School quality: Average                       | -0.015      | (0.021)        |
| School quality: Bad                           | -0.027      | (0.037)        |
| Distance to secondary school (km)             | -0.042***   | (0.005)        |
| Distance × Average school quality (Primary)   | -0.036**    | (0.017)        |
| Distance × Bad school quality (Primary)       | 0.057       | (0.046)        |
| Distance × Average school quality (Secondary) | 0.035***    | (0.013)        |
| Distance × Bad school quality (Secondary)     | -0.006      | (0.035)        |

*Note:*

Dependent variable: log house price.

Reference category for school quality: Good.

\*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

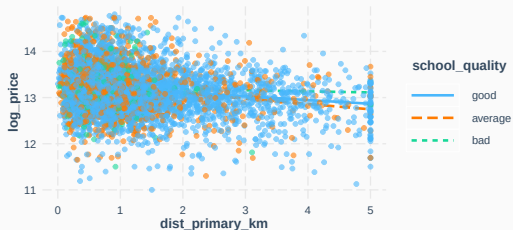
# Robustness check: Queen contiguity (Social Index)

| Variable                             | No interaction (Primary) | No interaction (Secondary) | With interaction (Primary) | With interaction (Secondary) |
|--------------------------------------|--------------------------|----------------------------|----------------------------|------------------------------|
| Social index                         | 0.001                    | 0.003                      | 0                          | 0.011*                       |
| Social index $\times$ Queen distance | NA                       | NA                         | 0.001                      | -0.005*                      |

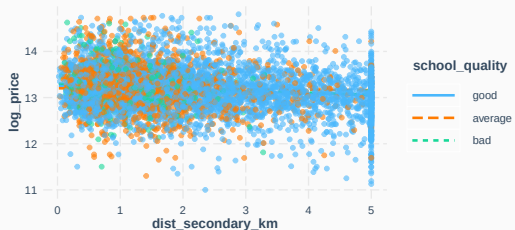
*Note:* Dependent variable: log house price. Control variables included but not shown. \*\*\*  $p < 0.01$ , \*\*  $p < 0.05$ , \*  $p < 0.1$

# Simple Slopes

Simple Slopes Analysis for 'dist\_primary\_km'  
by School Quality



Simple Slopes Analysis for 'dist\_secondary\_km'  
by School Quality



# Social Index: Interpretation of the Results

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## Primary School Proximity

- *Distance to primary school* is still *significant* and negatively affects house prices.
- Interaction Effect: The effect is *stronger for average schools* (Estimate = -0.041,  $p = 0.009$ ), suggesting that the price drop is greater when the school quality is average.
- *No significant effect* for bad schools ( $p = 0.2996$ ).

# Social Index: Interpretation of the Results

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## Secondary School Proximity

- *Distance to secondary school* is also still *significant* and negatively impacts house prices.
- The effect is *stronger for average schools* (Estimate = 0.0307,  $p = 0.0096$ ), meaning proximity to secondary schools is more impactful in areas with average school quality.
- *No significant effect* for bad schools ( $p = 0.7414$ ).



## **Concluding Thoughts and Limitations**

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# Concluding thoughts: Hypotheses Assessment

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**H1: House prices are higher related to schools**

*Expected: Negative coefficients on distance variables*

**Supported**

**H2: Social index moderates the effect of school proximity on house prices**

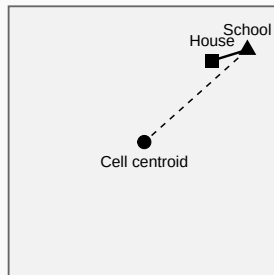
*Expected: Better schools leading to stronger price premiums*

**Not supported**

# Limitation: Direct Distance

The distance is not measured from the houses to the nearest school directly but from the school to the center of the raster cell.

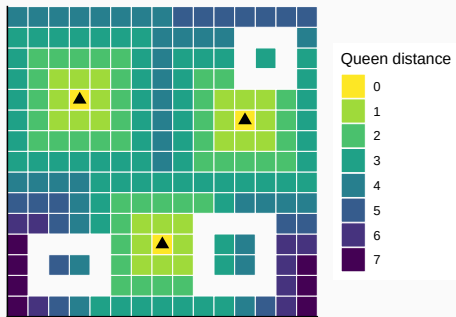
Single raster cell used for distance computation



# Limitation: Fragmented Grid

We do not have data for every grid cell of NRW but only a fragmented set of grids containing houses from the available datasets.

Queen distance to nearest school on a 15×15 raster grid



## Limitations: Other

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- **Listing prices:** The prices we have are not transaction but listing prices.

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